

Unlimited Vision: Professional Composition by Yourself

Xin Jin

jinxinbesti@foxmail.com

Beijing Electronic Science and Technology Institute
Beijing, China

Longteng Jiang

jang_lt@qq.com

Beijing Electronic Science and Technology Institute
Beijing, China

Liaoruxing Zhang

zhangxingxing002@gmail.com

Beijing Electronic Science and Technology Institute
Beijing, China

Dandan Li

lidd9432@gmail.com

Xidian University
Xi'an, China

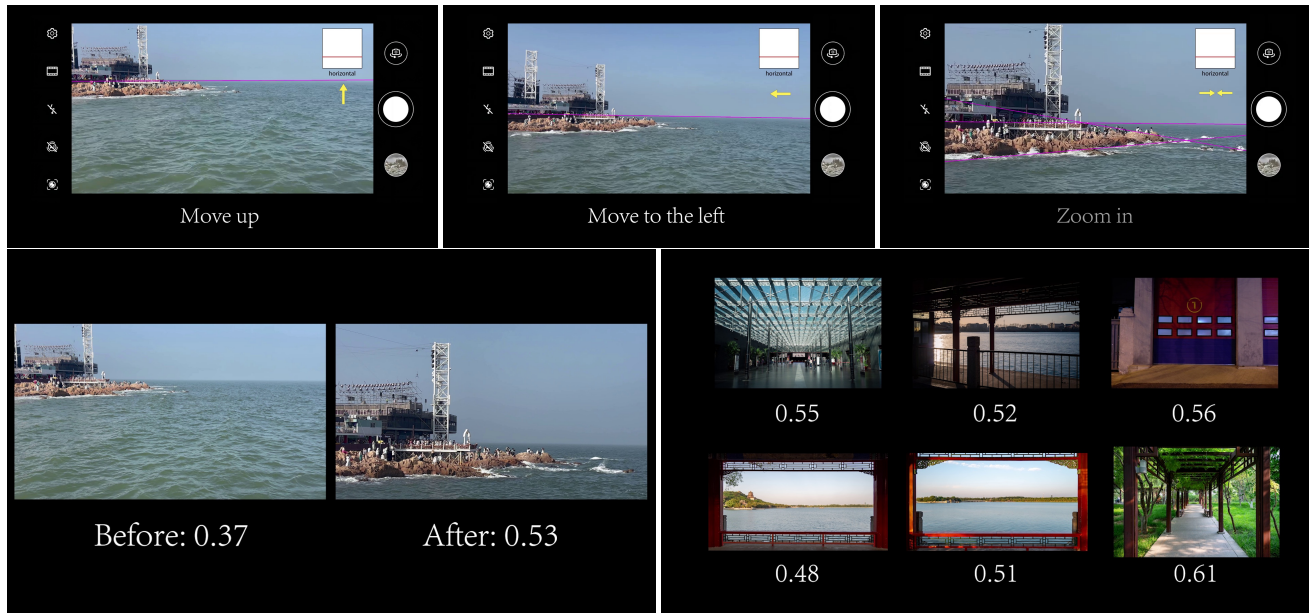


Figure 1: Our approach uses composition rules to guide an RT-DETR model, combining feature expansion to predict framing beyond image boundaries, enhancing real-time image composition for aesthetically superior photographs. The top row shows the guidance process for adjusting the viewpoint: moving up, left, and zooming in. The bottom left compares images before and after adjustment with their scores (0.37 and 0.53). The bottom right displays photos captured using our method with scores.

ABSTRACT

This paper introduces a novel method for enhancing image composition guidance in photography. It utilizes advanced composition rules to guide a Real-Time Detection Transformer (RT-DETR) model in predicting aesthetically pleasing compositions for photographs. Unlike traditional methods constrained by original image boundaries, our approach allows the predicted framing to extend beyond these limits, offering dynamic, real-time guidance for image composition in photography. The system integrates multi-label composition classification and compositional element annotation, using

YOLOv8 for key object detection and an enhanced Deep Hough Transform for compositional lines to guide photographers. It provides photographers with real-time guidance for optimal camera adjustments, transforming traditional post-processing tasks into an intuitive, interactive process. This method significantly enhances photographers' flexibility and effectiveness in capturing visually superior photographs.

CCS CONCEPTS

• Applied computing → Media arts.

KEYWORDS

photography guidance, aesthetic guidance, image aesthetics, photo composition rules, aesthetic evaluation

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

MM '24, October 28–November 1, 2024, Melbourne, VIC, Australia

© 2024 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0686-8/24/10

<https://doi.org/10.1145/3664647.3685001>

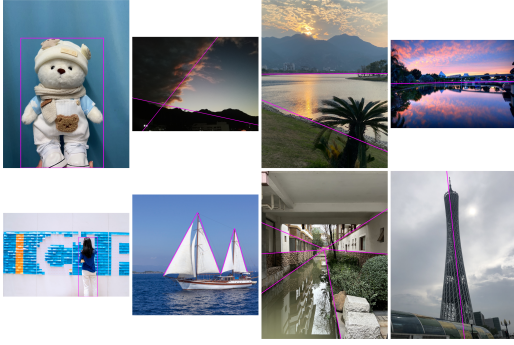


Figure 2: From left to right, the first row displays Center, Diagonal, Horizontal, and Symmetric compositions, while the second row features Rule of Thirds, Triangle, Vanishing Point, and Vertical compositions. Each image is annotated to highlight key compositional lines and objects, showcasing the application of these aesthetic principles.

1 INTRODUCTION

In professional photography, image composition is crucial for significantly influencing a photograph’s aesthetic appeal. Traditional methods for enhancing image composition have relied on post-processing techniques, such as image cropping [2, 3, 7, 10], to remove superfluous content and emphasize visually compelling elements. Despite their utility, these approaches often operate within the original image boundaries, limiting their effectiveness when the ideal composition extends beyond these borders.

This paper introduces a novel approach that utilizes composition rules (e.g., rule of thirds, diagonal composition) to guide a Real-Time Detection Transformer (RT-DETR)[9] model in predicting well-composed images. Unlike traditional methods, our approach allows the predicted framing to extend beyond the original image boundaries, overcoming the constraints of conventional image cropping. By predicting a compositionally optimal bounding box, our method guides users to adjust their camera’s position and zoom in real-time, enhancing the capture of aesthetically superior photographs.

Our contributions are threefold:

We propose a method that integrates composition principles into photographic guidance, enabling frame predictions beyond the original image boundaries. This approach overcomes the limitations of traditional image cropping.

We present a real-time interactive framework that suggests camera adjustments based on predictive framing. This helps photographers capture better-composed images.

Our framework supports various composition types with detailed annotations, enhancing the diversity and applicability of the guidance. These annotations provide better interpretability for users.

2 METHOD AND SYSTEM

2.1 Composition Classification and Annotation

Composition Classification In the field of image composition classification, we enhance the capabilities of [4] by adopting a

multi-label classification method. This approach is chosen because a single image can embody multiple composition styles simultaneously. Additionally, many exemplary photographic works integrate various composition techniques simultaneously, as described in [1]. Our method aims to emulate and accurately predict the composition styles of images. The classification includes eight conventional composition styles: center, diagonal, horizontal, rule of thirds (RoT), symmetric, triangle, vanishing point, and vertical.

Composition Element Annotation In the module for annotating compositional elements, we classify composition rules into two main forms of annotation: compositional lines and key object boxes. Key objects are annotated in real-time using YOLOv8, with two types of composition - Center composition and Rule of Thirds composition. For compositional lines, we enhance the Deep Hough Transform[8], with six types identified: Diagonal composition, Horizontal composition, Symmetric composition, Triangle composition, Vanishing Point composition, and Vertical composition. By filtering detected semantic lines based on specific compositional characteristics, symmetry axes and primary compositional lines are effectively captured. All annotations are shown in Figure2.

2.2 Photography Guidance

Optimal Composition Boundary Box Prediction By utilizing predicted composition rules and element annotations, our approach directs the RT-DETR model to predict composition frames that adhere to widely accepted aesthetic standards. By using [5], This enables real-time prediction of optimal composition boundary boxes that may extend beyond the original image boundaries. Guided by these rules and annotations, the model can explore potential compositions beyond the initial image constraints, enhancing the flexibility and creativity of the composition process.

Real-time Camera Adjustment Guidance The predicted composition frames provide actionable guidance for real-time camera adjustments. Users are given suggestions to reposition or zoom the camera to capture the most aesthetically pleasing images. This process includes aligning the camera’s center with the predicted view through vertical or horizontal translation, and then making zoom adjustments to match the predicted composition size. This interactive method transforms the conventional post-processing task of image cropping into a dynamic, real-time process, greatly enhancing the user’s photographic experience.

2.3 System Implementation

Our photography guidance user interface (UI) is crafted to be both intuitive and informative. In the top right corner, the UI displays the composition types that the current image conforms to. It also provides actionable instructions, suggesting necessary adjustments to the camera’s position and zoom to achieve optimal composition. Additionally, the UI annotates composition elements in real-time, helping users understand the rationale behind the suggested adjustments. This real-time feedback mechanism ensures users can effortlessly adhere to composition guidelines, resulting in aesthetically pleasing photographs. As illustrated in the figure1, these are examples of well-composed photographs captured using our method, with aesthetic scores annotated according to [6].

REFERENCES

- [1] David Bate. 2020. *Photography: the key concepts*. Routledge.
- [2] Guanjun Guo, Hanzi Wang, Chunhua Shen, Yan Yan, and Hong-Yuan Mark Liao. 2018. Automatic image cropping for visual aesthetic enhancement using deep neural networks and cascaded regression. *IEEE Transactions on Multimedia* 20, 8 (2018), 2073–2085.
- [3] Chaoyi Hong, Shuaiyuan Du, Ke Xian, Hao Lu, Zhiguo Cao, and Weicai Zhong. 2021. Composing photos like a photographer. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 7057–7066.
- [4] Jun-Tae Lee, Han-Ul Kim, Chul Lee, and Chang-Su Kim. 2018. Photographic composition classification and dominant geometric element detection for outdoor scenes. *Journal of Visual Communication and Image Representation* 55 (2018), 91–105.
- [5] Xiaoyu Liu, Ming Liu, Junyi Li, Shuai Liu, Xiaotao Wang, Lei Lei, and Wangmeng Zuo. 2023. Beyond Image Borders: Learning Feature Extrapolation for Unbounded Image Composition. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 13023–13032.
- [6] Hao Lou, Heng Huang, Chaoen Xiao, and Xin Jin. 2021. Aesthetic evaluation and guidance for mobile photography. In *Proceedings of the 29th ACM International Conference on Multimedia*. 2780–2782.
- [7] Bo Zhang, Li Niu, and Liqing Zhang. 2021. Image composition assessment with saliency-augmented multi-pattern pooling. *arXiv preprint arXiv:2104.03133* (2021).
- [8] Kai Zhao, Qi Han, Chang-Bin Zhang, Jun Xu, and Ming-Ming Cheng. 2021. Deep hough transform for semantic line detection. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 44, 9 (2021), 4793–4806.
- [9] Yian Zhao, Wenyu Lv, Shangliang Xu, Jinman Wei, Guanzhong Wang, Qingqing Dang, Yi Liu, and Jie Chen. 2023. Detsr beat yolos on real-time object detection. *arXiv preprint arXiv:2304.08069* (2023).
- [10] Lei Zhong, Feng-Heng Li, Hao-Zhi Huang, Yong Zhang, Shao-Ping Lu, and Jue Wang. 2021. Aesthetic-guided outward image cropping. *ACM Transactions on Graphics (TOG)* 40, 6 (2021), 1–13.